

Classical Mechanics: Newton's Three Laws

1. First Law — Inertia

An object remains at rest, or in uniform motion, unless acted on by a net external force.

This defines what a force is: something that changes velocity, not something needed to sustain it.

2. Second Law — $F = ma$

Net force equals mass times acceleration. Acceleration is directly proportional to force and inversely proportional to mass.

- Doubling the force doubles the acceleration (mass fixed).
- Doubling the mass halves the acceleration (force fixed).
- Always resolve forces into components before applying $F = ma$ along an axis.

3. Third Law — Action & Reaction

For every force A exerts on B, B exerts an equal and opposite force on A. These act on different bodies, so they never cancel each other out.

4. Worked Example

A 5 kg block on a frictionless surface is pushed with 20 N. Acceleration = $F/m = 20/5 = 4 \text{ m/s}^2$.